

AI ENGINEERING COHORT · DELIVERABLE 4

Architecture

Conversational Memory Intelligence System — Diagrams

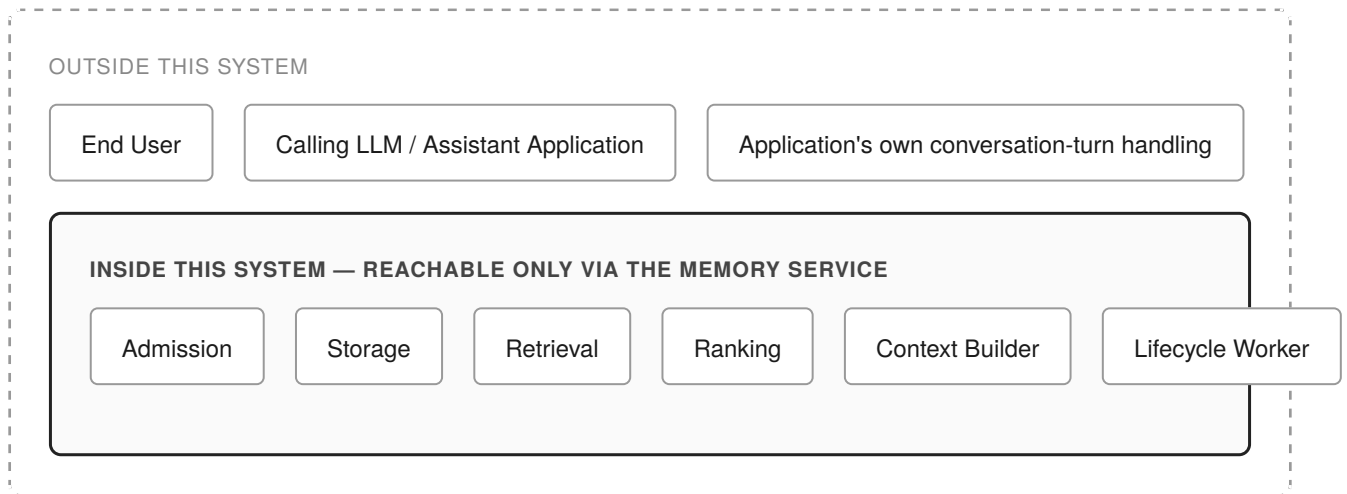
Abhijeet Hiwale

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design/architecture.pdf

1. System Boundary

Expands system_design.pdf, Section 1. What's inside this memory system vs. what belongs to the calling application.

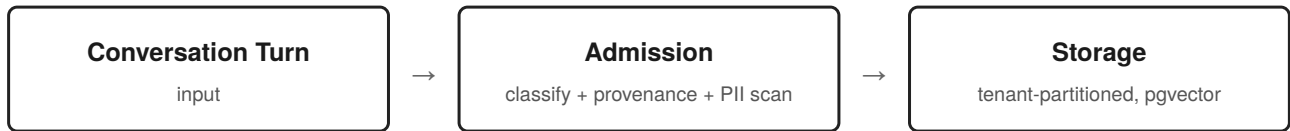


The calling application sends conversation turns in and a query out. It never addresses an internal component directly — every request passes through the Memory Service.

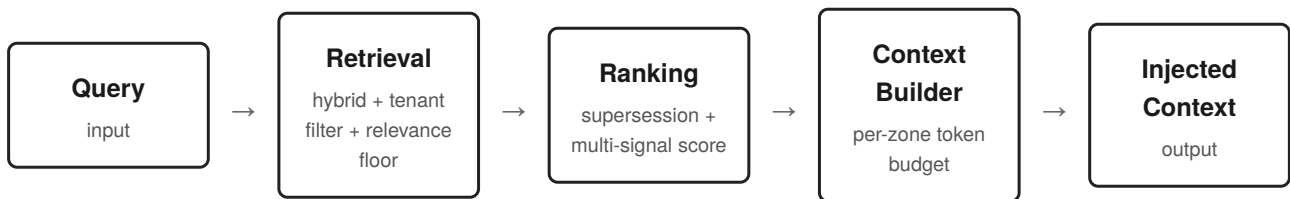
2. Component Flow — Online Path

Expands system_design.pdf, Section 5 & 8. The synchronous request path — every box here is on a real, enforced latency budget (system_design.pdf, Section 3).

WRITE FLOW — TRIGGERED BY A NEW CONVERSATION TURN



READ FLOW — TRIGGERED BY A QUERY NEEDING MEMORY CONTEXT



BACKGROUND PATH — ASYNCHRONOUS, OFF THE REQUEST PATH



Write and read flows are independent operations against the same Storage component, not steps in one pipeline — a slow admission does not block a concurrent query's retrieval latency budget (system_design.pdf, Section 8).

3. Component Responsibility Reference

Expands `system_design.pdf`, Section 6. Quick-reference boundary table — what each component owns and explicitly does not.

COMPONENT	RESPONSIBLE FOR	NOT RESPONSIBLE FOR
Admission	ADD/UPDATE/DELETE/NOOP classification, provenance tagging, PII pre-filter	Ranking, relevance judgments
Storage	Persistence, tenant partitioning, validity-window fields	Similarity scoring, relevance
Retrieval	Hybrid search, relevance floor, tenant pre-filter	Conflict resolution (that's Ranking)
Ranking	Supersession, low-confidence fallback, final ordering	Storage, admission-time filtering
Context Builder	Token budgeting, zone allocation, truncation	Which memories were retrieved
Lifecycle Worker	Decay, merge, eviction, deletion propagation	Real-time retrieval-path decisions

Full narrative detail for every box on this page is in `system_design.pdf`, Sections 5–9. This document is the visual index into that text, not a replacement for it.